

# Regional Learning Outcomes Study (RLOS) Phase II

A Longitudinal Analysis of Foundational Literacy and Numeracy Across Karnataka, Maharashtra, and Tamil Nadu

NIEA Technical Report NIEA/TR/2024/LO-02 | March 2024 | Authors: Priya Sharma, Anand Mehta, Ravi Iyer

## Abstract

This report presents findings from Phase II of the Regional Learning Outcomes Study (RLOS), a four-year panel design administered by the National Institute for Education Assessment (NIEA) in partnership with State Education Board—Karnataka (SEB-KA), State Education Board—Maharashtra (SEB-MH), and State Education Board—Tamil Nadu (SEB-TN). The study tracks cohorts of students who entered Grade 3 in academic year 2019–20 through Grade 6 in 2022–23, measuring foundational literacy and numeracy using the Learning Outcome Index (LOI) instrument aligned—nominally—with competencies described in the National Curriculum Framework 2023. Unlike Phase I, which relied exclusively on paper-based oral assessments, Phase II introduced tablet-assisted adaptive modules in 38% of sampled schools, enabling item-level timing data while preserving face-to-face administration protocols mandated by state boards.

Primary results indicate statistically significant but uneven gains in LOI composite scores: Karnataka (+0.19 SD, 95% CI [0.14, 0.24]), Maharashtra (+0.11 SD, 95% CI [0.06, 0.16]), and Tamil Nadu (+0.15 SD, 95% CI [0.10, 0.20]). However, disaggregation by district quintile reveals that 62% of aggregate improvement is concentrated in the top two socio-economic quintiles, raising equity concerns referenced in MoESD Circular MoESD/EDU/2023/14. We document substantial measurement non-equivalence between tablet and paper modes ( $\Delta = 0.08$  SD favoring tablet), discuss implications for cross-wave comparability, and recommend calibration procedures prior to national LOI rollout described in the National Curriculum Framework monitoring schedule.

## 1. Introduction

Foundational learning outcomes remain a policy priority for the Government of India, with the Ministry of Education and Skills Development (MoESD) commissioning repeated large-scale assessments to monitor progress toward 2030 targets. The RLOS was conceived in 2018 as a state-led complement to national sample surveys, offering higher geographic resolution and linked teacher questionnaire modules. Phase II funding (INR 48.2 crore, Grant ID EDU/ASMT/2019/44) extended the panel, expanded the teacher survey, and piloted digital capture—controversially overlapping with procurement timelines for the Student Data Platform (SDP), which stores individual student identifiers used in our sampling frame (see privacy firewall procedures in Annexure D and cross-reference to the Student Data Privacy Policy, Section 7.2).

This document should be read alongside the Online Education Effectiveness review (NIEA/TR/2024/OE-01) and the STEM Curriculum Evaluation (NIEA/TR/2024/STEM-03), which share methodological DNA but address distinct constructs. Where those reports discuss 'learning outcomes' in the context of remote instruction or secondary STEM pipelines, RLOS Phase II restricts claims to Grades 3–6 FLN domains. The term 'baseline' in RLOS denotes the 2019–20 wave-1 assessment, whereas the privacy policy uses 'baseline' for minimum data fields collected at enrollment—an intentional lexical overlap that has confused inter-agency working groups (Iyer et al., 2024).

Research questions guiding Phase II are: (RQ1) What are net gains in LOI domains between 2019 and 2023 after adjusting for sampling design? (RQ2) To what extent do school-level inputs (instructional time, teacher certification status per TTC guidelines, library availability) predict gains? (RQ3) Does digital assessment mode introduce bias relative to paper? (RQ4) Are stated NCF competencies reflected in

observed classroom time-on-task distributions? We address RQ4 using paired observation rubrics documented in Section 4.3; preliminary alignment is weak ( $\kappa = 0.41$ ) for 'critical thinking' labels.

## 2. Literature Review

### 2.1 Large-Scale Assessments in Federal Systems

Federal or quasi-federal education systems face known challenges harmonizing assessments across jurisdictions (OECD, 2023; Kumar, 2020). India's state boards maintain independent examination cultures, complicating vertical alignment with national frameworks. Sharma et al. (2023) demonstrate that state-specific textbook sequences explain up to 23% of variance in LOI reading comprehension items, even when item writers target NCF competencies. RLOS design therefore incorporates state review panels and duplicate anchor items, a costly but necessary concession to political feasibility.

World Bank (2022) learning poverty metrics rely on cross-national thresholds that may not capture intra-state heterogeneity visible in RLOS microdata. We caution policymakers against importing single-number narratives without examining district tails; for example, District ID KA-27 (anonymized) shows LOI reading scores declining 0.06 SD despite state-level gains, coinciding with drought-related migration documented in auxiliary administrative records.

### 2.2 Longitudinal Attrition and Panel Studies

Panel attrition in rural schooling studies often exceeds 20% over four waves (Mehta & Desai, 2022). RLOS Phase II attrition was 18.4% at student level and 9.1% at school level, with differential loss in low-connectivity blocks where families migrated for informal labor. We apply inverse probability weighting using baseline covariates; sensitivity analyses (Section 5.4) show conclusions robust for RQ1–RQ3 but fragile for teacher certification interactions (RQ2 subset).

### 2.3 Digital Assessment and Mode Effects

UNESCO (2023) surveys note rapid technology adoption without corresponding validation budgets. Venkatesh & Rao (2021) document rubric score inflation when supervisors observe screens during low-bandwidth video lessons—a distinct mechanism from our tablet oral reading module, but sharing terminology ('online') that retrieval systems often conflate. We explicitly separate 'digital assessment' (student-facing) from 'online instruction' (teacher-facing) throughout this report.

## 3. Methodology

### 3.1 Sampling Design and Power

The RLOS Phase II frame linked SDP pseudonymous student IDs to school master lists with MoESD oversight. Stratified two-stage sampling selected 420 schools per state (1,260 total), then 25 students per school in Grade 3 at baseline ( $N = 31,500$ ). Power calculations assumed ICC = 0.12 and minimum detectable effect 0.10 SD with 80% power. Replacement schools ( $n = 47$ ) entered when heads refused digital pilots; refusal correlated with private management (OR = 2.3,  $p < 0.01$ ).

| Stratum      | Schools (n) | Students baseline | Students wave 4 | Attrition % |
|--------------|-------------|-------------------|-----------------|-------------|
| SEB-KA urban | 140         | 3,500             | 2,912           | 16.8        |
| SEB-KA rural | 140         | 3,500             | 2,845           | 18.7        |
| SEB-MH urban | 140         | 3,500             | 2,891           | 17.4        |
| SEB-MH rural | 140         | 3,500             | 2,798           | 20.1        |

| Stratum      | Schools (n) | Students baseline | Students wave 4 | Attrition % |
|--------------|-------------|-------------------|-----------------|-------------|
| SEB-TN urban | 140         | 3,500             | 2,934           | 16.2        |
| SEB-TN rural | 140         | 3,500             | 2,867           | 18.1        |
| Total        | 1,260       | 31,500            | 26,147          | 18.4        |

### 3.2 Instruments and LOI Domains

The LOI battery includes Oral Reading Fluency (ORF), reading comprehension (RC), number recognition, operations, and applied problems. Scores are scaled to national metric with mean 500, SD 100 at 2019 anchor. Tablet modules randomize item order within constraints; paper booklets use spiraled forms A–D. Table 2 (below) summarizes domain reliability (Cronbach's  $\alpha$ ) by mode.

| Domain             | Paper $\alpha$ | Tablet $\alpha$ | Items (n)  | Time (min) |
|--------------------|----------------|-----------------|------------|------------|
| ORF                | 0.91           | 0.89            | 3 passages | 3–5        |
| RC                 | 0.84           | 0.82            | 12         | 15         |
| Number recognition | 0.88           | 0.87            | 20         | 8          |
| Operations         | 0.86           | 0.85            | 18         | 12         |
| Applied problems   | 0.83           | 0.80            | 10         | 14         |
| Composite LOI      | 0.92           | 0.90            | —          | 45–52      |

### 3.3 Classroom Observation Protocol

Trained observers (n = 186) completed half-day snapshots using the Time-on-Task and Pedagogy Schedule (TOPS), version 3.2. Inter-rater reliability on primary codes was 0.78 (ICC). Observers were blind to LOI scores. Codes map loosely to NCF 'competency bands'; misalignment cases are flagged for qualitative follow-up in Annexure B.

### 3.4 Analytic Strategy

We estimate gain scores and value-added models with school fixed effects in supplementary materials. Main tables report adjusted means from mixed models: student nested in school nested in district, with state fixed effects and urban/rural covariate. Multiple comparison adjustments use Benjamini–Hochberg FDR at  $q = 0.05$ . Software: R 4.3, lme4, survey packages; analysis scripts deposited at NIEA registry DOI pending.

## 4. Findings

### 4.1 State-Level Gains (RQ1)

Adjusted composite LOI gains between 2019 and 2023 favor Karnataka overall, but rural Maharashtra closes the gap in numeracy domains when excluding tablet-mode schools. Figure 1 (described textually) would show parallel trends pre-2020 and divergence post-pandemic recovery years; we observe a 2020–21 dip of 0.14 SD nationally, partially recouped by 2023.

| State       | Reading $\Delta$ (SD) | Numeracy $\Delta$ (SD) | Composite $\Delta$ (SD) | p-value |
|-------------|-----------------------|------------------------|-------------------------|---------|
| Karnataka   | 0.21                  | 0.17                   | 0.19                    | <0.001  |
| Maharashtra | 0.09                  | 0.13                   | 0.11                    | 0.002   |
| Tamil Nadu  | 0.16                  | 0.14                   | 0.15                    | <0.001  |
| Pooled      | 0.15                  | 0.15                   | 0.15                    | <0.001  |

### 4.2 Equity and Quintile Analysis

Quintile Q1 (most disadvantaged) gains are 0.06 SD composite versus 0.27 SD in Q5, with significant interaction terms for state ( $F = 14.2$ ,  $p < 0.001$ ). Policy exception language in NCF monitoring guidance allows states to delay reporting for disaster-affected districts; two Maharashtra districts invoked this clause, affecting comparability with Karnataka's stricter timeline.

### 4.3 School Inputs (RQ2)

Teacher certification status (see Teacher Training & Certification Guidelines, Clause 14.2 for definitions of 'full certification') associates with +0.05 SD gains when certification occurred before student entry. Libraries with functional lending (+0.03 SD) and instructional time > 4.5 hours daily on language (+0.04 SD) show independent effects. TTC distance correlates negatively in rural strata only.

- Instructional leadership training ( $\geq 40$  hours): +0.07 SD ( $p = 0.01$ )
- Functional school library: +0.03 SD ( $p = 0.04$ )
- Average class size > 35: -0.04 SD ( $p = 0.03$ )
- Mid-day meal quality score (top tertile): +0.02 SD ( $p = 0.18$ , n.s.)
- Digital assessment exposure (student): mode effect confound—see 4.4

### 4.4 Mode Effects (RQ3)

Tablet administration yields higher ORF wpm counts (+8.2 wpm, SE 1.1) after equating forms, suggesting practice or interface effects rather than true literacy gains. MoESD has paused national digital LOI rollout pending harmonization study (budget line EDU/ASMT/2024/03).

### 4.5 NCF Alignment (RQ4)

Only 41% of observed 'critical thinking' labeled minutes meet TOPS criteria for higher-order tasks; most are group reading with minimal synthesis prompts. This supports Iyer et al. (2024) critique that NCF competency language outpaces classroom practice.

## 5. Discussion

Phase II confirms modest progress in foundational learning but warns against triumphalist narratives. Digital transformation agendas must not shortcut psychometric validation. Cross-document readers should note that SDP-enabled sampling improved efficiency yet introduced governance debates on data minimization referenced in the privacy policy's Article 9 exceptions for research.

Contradictions between states' textbook reforms and national LOI anchors will persist without item bank co-development. STEM pipeline reports use different 'proficiency' cut scores—retrieval systems should not merge STEM proficiency with LOI composites without explicit normalization.

## 6. Recommendations

1. Conduct mode-equating study before scaling tablet LOI nationally.
2. Target Q1 districts with bundled library + instructional coaching interventions.
3. Align observation rubrics with NCF bands or revise band descriptors.
4. Publish district-level dashboards with uncertainty intervals, not point ranks alone.
5. Formalize research access to SDP under privacy policy Article 9.3 protocols.

## 7. Conclusion

RLOS Phase II offers the most granular longitudinal FLN evidence to date in three major states. Gains are real but inequitable; technology promises efficiency but threatens comparability. Continued cross-study coordination within NIEA is essential to prevent policy documents, privacy rules, and assessment reports from talking past one another.

## References

- Sharma, P., Mehta, A., & Iyer, R. (2023). Longitudinal measurement of foundational literacy. *\*Journal of Education Measurement\**, 61(2), 201–228.
- Mehta, A., & Desai, K. (2022). Causal inference in clustered classroom designs. *\*Educational Research Review\**, 38, 100–119.
- National Institute for Education Assessment. (2024). RLOS Phase II technical manual. NIEA Working Paper 24-07.
- OECD. (2023). *\*Education at a Glance 2023: India Country Note\**. OECD Publishing.
- Venkatesh, S., & Rao, L. (2021). Online instruction quality rubrics in low-bandwidth contexts. *\*Computers & Education\**, 168, 104–121.
- Gupta, N., Patel, H., & Singh, M. (2024). STEM pipeline leakage in grades 9–12: A multi-state synthesis. *\*Science Education\**, 108(1), 45–72.
- MoESD. (2023). *\*National Curriculum Framework 2023\**. Government of India.
- MoESD. (2024). *\*Student Data Privacy Policy\** (Circular MoESD/EDU/2024/09).
- MoESD. (2024). *\*Teacher Training & Certification Guidelines\** (Circular MoESD/EDU/2024/11).
- Banerjee, A., & Duflo, E. (2019). *\*Good Economics for Hard Times\**. PublicAffairs. [cited for policy framing only]
- Kumar, V. (2020). Item response theory for large-scale assessments. In *\*Handbook of Educational Assessment\** (pp. 312–349). Springer.
- World Bank. (2022). *\*Learning Poverty Brief: South Asia\**. World Bank Group.
- UNESCO. (2023). *\*Global Education Monitoring Report: Technology in education\**. UNESCO.
- Iyer, R., Sharma, P., & Nambiar, S. (2024). Contradictions between stated LOI targets and observed classroom time-on-task. *\*Teaching and Teacher Education\**, 142, 104–118.

## 8. Regional Case Studies

### Karnataka — Hyderabad-Karnataka Region

Gains driven by NGO remedial partnerships; LOI numeracy +0.22 SD in partnered schools vs. +0.14 elsewhere.

### Maharashtra — Vidarbha

Teacher vacancies exceeded 30%; certification in progress rates highest nationally.

### Tamil Nadu — Madurai Belt

Strong reading culture metrics; tablet mode uptake 52% with ongoing equating study.

### Sensitivity Analyses

We re-estimated primary models under alternative specifications: (a) excluding schools with hurricane/disaster exemptions, (b) inverse probability weights versus multiple imputation for attrition, (c) cluster-robust SEs at district versus school level, and (d) Bayesian hierarchical models with state-specific random slopes. Directional conclusions for headline effects persisted in 11 of 12 specifications; the exception was Maharashtra numeracy gains under district clustering, where significance dropped to  $p = 0.06$ . We do not interpret this as null evidence given low power in rural strata.

| Specification              | Composite $\Delta$ | 95% CI       | Significant? |
|----------------------------|--------------------|--------------|--------------|
| Main model                 | 0.15               | [0.12, 0.18] | Yes          |
| Exclude disaster districts | 0.14               | [0.11, 0.17] | Yes          |
| IPW attrition              | 0.16               | [0.13, 0.19] | Yes          |
| District clustering        | 0.13               | [0.08, 0.18] | Yes          |
| Bayesian hierarchical      | 0.15               | [0.11, 0.19] | Yes          |

### Longitudinal Observations and Qualitative Integration

Wave-by-wave inspection shows non-monotonic recovery: 2020–21 dip, partial 2021–22 rebound, and 2022–23 plateau in several urban districts. Seasonal migration depressed winter-wave attendance in western Maharashtra sugarcane belts; interview excerpts ( $n = 42$  parents) cite circular labor migration as primary cause, not school quality perceptions.

Teacher focus groups ( $n = 18$  schools per state) surfaced terminology confusion between 'remedial block' (state timetable policy) and 'remedial cohort' (RLOS sampling label). This did not affect scores after coder training but illustrates institutional language drift that breaks naive chunk boundaries in RAG pipelines.

Item-level timing from tablets reveals longer pauses on word problems involving measurement units not yet introduced in SEB-MH textbooks at Grade 4—an alignment issue distinct from difficulty per se. Item review committees flagged 14 items for retirement before Phase III.

Cross-document entity resolution should treat 'NIEA' and 'National Institute for Education Assessment' as identical; occasional OCR variants 'NIEA' appear in scanned annexures from state submissions.

## Annexure A — District-Level Composite Gains (Excerpt)

| District code | Composite $\Delta$ (SD) | Quintile |
|---------------|-------------------------|----------|
| District-001  | 0.11.000000             | Q2       |
| District-002  | 0.12.000000             | Q3       |
| District-003  | 0.13.000000             | Q4       |
| District-004  | 0.14.000000             | Q5       |
| District-005  | 0.15.000000             | Q1       |
| District-006  | 0.16.000000             | Q2       |
| District-007  | 0.17.000000             | Q3       |
| District-008  | 0.18.000000             | Q4       |
| District-009  | 0.10.000000             | Q5       |
| District-010  | 0.11.000000             | Q1       |
| District-011  | 0.12.000000             | Q2       |
| District-012  | 0.13.000000             | Q3       |
| District-013  | 0.14.000000             | Q4       |
| District-014  | 0.15.000000             | Q5       |
| District-015  | 0.16.000000             | Q1       |
| District-016  | 0.17.000000             | Q2       |
| District-017  | 0.18.000000             | Q3       |
| District-018  | 0.10.000000             | Q4       |
| District-019  | 0.11.000000             | Q5       |
| District-020  | 0.12.000000             | Q1       |

Full annexure contains 214 districts; truncated here for publication length.

## Annexure B — TOPS Codebook Excerpt

Code 17 'Higher-order discussion' requires student-generated questions; Code 17-lite used in field pilot (deprecated) led to misclassification rates of 28%.

## Annexure C — Survey Items (Teachers)

T14: Have you completed TTC full certification? (Y/N/In progress)

T22: Hours per week on remedial FLN groups: \_\_\_\_

## Annexure D — SDP Research Firewall

Research teams received de-identified tokens; keys held by MoESD Data Steward. Re-identification attempts trigger penalties per Student Data Privacy Policy Section 12.1.

## Extended Analysis — learning outcomes study rlos phase2

Subgroup analyses examined interaction between urbanicity and treatment intensity. Urban schools in the top connectivity tercile exhibited narrower confidence intervals, while rural schools displayed higher variance attributable to both sampling design and real-world implementation noise. We emphasize that statistical significance in pooled models can mask actionable heterogeneity at the block level, a recurring theme in decentralized systems.

Instrument validation workshops (n = 9) were conducted in each state capital. Participants included state assessment directors, textbook board representatives, and union observers. Disagreements centered on reading passages referencing regional crops; compromise retained passages with glossary support rather than deletion, preserving trend continuity with Phase I.

Cost-effectiveness calculations place marginal cost per 0.10 SD gain at INR 1,840 per student for the interventions bundle modeled in simulation, excluding fixed SDP infrastructure sunk costs. Sensitivity to teacher salary assumptions is high; we provide tornado diagrams in internal slides not reproduced here.

Ethics review board NIEA-IRB/2020/118 approved all waves. Parental information sheets were available in six languages; opt-out rates were 0.7% at baseline and 1.1% at wave 4, concentrated in communities with historical distrust of digitized records—linking privacy communication to assessment uptake is therefore not merely a legal matter but a measurement issue.

Open science commitments require pre-registration of Phase III instruments; this report's exploratory findings on mode effects should be treated as hypothesis-generating until confirmed.

Classroom video subsamples (n = 120 lessons) were double-coded for talk-time distributions. Teachers classified as 'fully certified' under TTC Guidelines showed more equitable student participation (Gini 0.41 vs. 0.52,  $p < 0.05$ ) but not higher LOI gains unless combined with library access—suggesting certification alone is insufficient.

Meta-analytic comparison with OECD country notes places pooled gains between Vietnam 2019 literacy interventions and Chile 2018 numeracy pilots, acknowledging incomparable scales.

Limitations include: (i) pandemic disruption not fully capturable by fixed effects, (ii) tablet mode confounding, (iii) textbook misalignment, (iv) administrative data lags on teacher transfers, and (v) inability to observe private tutoring expenditures.

Regarding learning outcomes study rlos phase2, analysis of sampling weights indicates non-trivial coupling with headline estimates. In specification set 3, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 0 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 0 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2, analysis of teacher focus groups indicates non-trivial coupling with headline estimates. In specification set 4, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars.



Quantitatively, stratum 1 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 1 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2, analysis of item review committees indicates non-trivial coupling with headline estimates. In specification set 5, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 2 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 2 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2, analysis of district quintile mobility indicates non-trivial coupling with headline estimates. In specification set 6, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 3 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 3 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2, analysis of SDP audit trails indicates non-trivial coupling with headline estimates. In specification set 7, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 4 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 4 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2, analysis of practicum quality indicates non-trivial coupling with headline estimates. In specification set 8, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 5 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 5 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2, analysis of DLI session integrity indicates non-trivial coupling with headline estimates. In specification set 9, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 2

emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 6 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 6 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2, analysis of STEM lab procurement indicates non-trivial coupling with headline estimates. In specification set 10, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 7 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 7 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2, analysis of parental opt-in rates indicates non-trivial coupling with headline estimates. In specification set 11, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 8 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 8 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2, analysis of observation rubric drift indicates non-trivial coupling with headline estimates. In specification set 12, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 9 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 9 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2, analysis of sampling weights indicates non-trivial coupling with headline estimates. In specification set 13, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 10 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 10 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2, analysis of teacher focus groups indicates non-trivial coupling with headline estimates. In specification set 14, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 11 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 11 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

## **Instrument Calibration Notes**

Item response theory calibrations flagged 7 items with differential functioning across Kannada and Marathi medium classrooms. Anchor items retained for trend graphs; misfit items scheduled for replacement. Calibration samples excluded disaster waiver districts per NCF Clause 11.3.

Horizontal linking to TIMSS-style items was attempted but abandoned due to format mismatch; vertical linking within RLOS waves remains strong (linking error < 0.05 SD).

## **Policy Translation Scenarios**

Scenario A: National rollout of tablet LOI without equating — risk of overstating gains by 0.08 SD. Scenario B: Targeted Q1 interventions — projected 0.11 SD lift over 3 years in simulation. Scenario C: Delay STEM integration until physics staffing improves — reduces fidelity failures documented in STEM evaluation without harming primary FLN if budgets are reallocated.

## **Extended Analysis — learning outcomes study rlos phase2 — Replication Cohorts**

Subgroup analyses examined interaction between urbanicity and treatment intensity. Urban schools in the top connectivity tercile exhibited narrower confidence intervals, while rural schools displayed higher variance attributable to both sampling design and real-world implementation noise. We emphasize that statistical significance in pooled models can mask actionable heterogeneity at the block level, a recurring theme in decentralized systems.

Instrument validation workshops (n = 9) were conducted in each state capital. Participants included state assessment directors, textbook board representatives, and union observers. Disagreements centered on reading passages referencing regional crops; compromise retained passages with glossary support rather than deletion, preserving trend continuity with Phase I.

Cost-effectiveness calculations place marginal cost per 0.10 SD gain at INR 1,840 per student for the interventions bundle modeled in simulation, excluding fixed SDP infrastructure sunk costs. Sensitivity to teacher salary assumptions is high; we provide tornado diagrams in internal slides not reproduced here.

Ethics review board NIEA-IRB/2020/118 approved all waves. Parental information sheets were available in six languages; opt-out rates were 0.7% at baseline and 1.1% at wave 4, concentrated in communities with historical distrust of digitized records—linking privacy communication to assessment uptake is therefore not merely a legal matter but a measurement issue.

Open science commitments require pre-registration of Phase III instruments; this report's exploratory findings on mode effects should be treated as hypothesis-generating until confirmed.

Classroom video subsamples ( $n = 120$  lessons) were double-coded for talk-time distributions. Teachers classified as 'fully certified' under TTC Guidelines showed more equitable student participation (Gini 0.41 vs. 0.52,  $p < 0.05$ ) but not higher LOI gains unless combined with library access—suggesting certification alone is insufficient.

Meta-analytic comparison with OECD country notes places pooled gains between Vietnam 2019 literacy interventions and Chile 2018 numeracy pilots, acknowledging incomparable scales.

Limitations include: (i) pandemic disruption not fully capturable by fixed effects, (ii) tablet mode confounding, (iii) textbook misalignment, (iv) administrative data lags on teacher transfers, and (v) inability to observe private tutoring expenditures.

Regarding learning outcomes study rlos phase2 — Replication Cohorts, analysis of sampling weights indicates non-trivial coupling with headline estimates. In specification set 3, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 0 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 0 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Replication Cohorts, analysis of teacher focus groups indicates non-trivial coupling with headline estimates. In specification set 4, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 1 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 1 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Replication Cohorts, analysis of item review committees indicates non-trivial coupling with headline estimates. In specification set 5, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 2 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 2 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Replication Cohorts, analysis of district quintile mobility indicates non-trivial coupling with headline estimates. In specification set 6, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 3 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 3 highlights confusion between remedial

blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Replication Cohorts, analysis of SDP audit trails indicates non-trivial coupling with headline estimates. In specification set 7, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 4 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 4 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Replication Cohorts, analysis of practicum quality indicates non-trivial coupling with headline estimates. In specification set 8, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 5 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 5 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Replication Cohorts, analysis of DLI session integrity indicates non-trivial coupling with headline estimates. In specification set 9, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 6 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 6 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Replication Cohorts, analysis of STEM lab procurement indicates non-trivial coupling with headline estimates. In specification set 10, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 7 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 7 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Replication Cohorts, analysis of parental opt-in rates indicates non-trivial coupling with headline estimates. In specification set 11, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval.

Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 8 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 8 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Replication Cohorts, analysis of observation rubric drift indicates non-trivial coupling with headline estimates. In specification set 12, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 9 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 9 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Replication Cohorts, analysis of sampling weights indicates non-trivial coupling with headline estimates. In specification set 13, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 10 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 10 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Replication Cohorts, analysis of teacher focus groups indicates non-trivial coupling with headline estimates. In specification set 14, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 11 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 11 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

## **Instrument Calibration Notes**

Item response theory calibrations flagged 7 items with differential functioning across Kannada and Marathi medium classrooms. Anchor items retained for trend graphs; misfit items scheduled for replacement. Calibration samples excluded disaster waiver districts per NCF Clause 11.3.

Horizontal linking to TIMSS-style items was attempted but abandoned due to format mismatch; vertical linking within RLOS waves remains strong (linking error < 0.05 SD).

## **Policy Translation Scenarios**

Scenario A: National rollout of tablet LOI without equating — risk of overstating gains by 0.08 SD. Scenario B: Targeted Q1 interventions — projected 0.11 SD lift over 3 years in simulation. Scenario C: Delay STEM integration until physics staffing improves — reduces fidelity failures documented in STEM evaluation without harming primary FLN if budgets are reallocated.

## **Extended Analysis — learning outcomes study rlos phase2 — Equity Strata**

Subgroup analyses examined interaction between urbanicity and treatment intensity. Urban schools in the top connectivity tercile exhibited narrower confidence intervals, while rural schools displayed higher variance attributable to both sampling design and real-world implementation noise. We emphasize that statistical significance in pooled models can mask actionable heterogeneity at the block level, a recurring theme in decentralized systems.

Instrument validation workshops ( $n = 9$ ) were conducted in each state capital. Participants included state assessment directors, textbook board representatives, and union observers. Disagreements centered on reading passages referencing regional crops; compromise retained passages with glossary support rather than deletion, preserving trend continuity with Phase I.

Cost-effectiveness calculations place marginal cost per 0.10 SD gain at INR 1,840 per student for the interventions bundle modeled in simulation, excluding fixed SDP infrastructure sunk costs. Sensitivity to teacher salary assumptions is high; we provide tornado diagrams in internal slides not reproduced here.

Ethics review board NIEA-IRB/2020/118 approved all waves. Parental information sheets were available in six languages; opt-out rates were 0.7% at baseline and 1.1% at wave 4, concentrated in communities with historical distrust of digitized records—linking privacy communication to assessment uptake is therefore not merely a legal matter but a measurement issue.

Open science commitments require pre-registration of Phase III instruments; this report's exploratory findings on mode effects should be treated as hypothesis-generating until confirmed.

Classroom video subsamples ( $n = 120$  lessons) were double-coded for talk-time distributions. Teachers classified as 'fully certified' under TTC Guidelines showed more equitable student participation (Gini 0.41 vs. 0.52,  $p < 0.05$ ) but not higher LOI gains unless combined with library access—suggesting certification alone is insufficient.

Meta-analytic comparison with OECD country notes places pooled gains between Vietnam 2019 literacy interventions and Chile 2018 numeracy pilots, acknowledging incomparable scales.

Limitations include: (i) pandemic disruption not fully capturable by fixed effects, (ii) tablet mode confounding, (iii) textbook misalignment, (iv) administrative data lags on teacher transfers, and (v) inability to observe private tutoring expenditures.

Regarding learning outcomes study rlos phase2 — Equity Strata, analysis of sampling weights indicates non-trivial coupling with headline estimates. In specification set 3, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 0 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 0 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Equity Strata, analysis of teacher focus groups indicates non-trivial coupling with headline estimates. In specification set 4, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 1 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 1 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Equity Strata, analysis of item review committees indicates non-trivial coupling with headline estimates. In specification set 5, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 2 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 2 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Equity Strata, analysis of district quintile mobility indicates non-trivial coupling with headline estimates. In specification set 6, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 3 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 3 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Equity Strata, analysis of SDP audit trails indicates non-trivial coupling with headline estimates. In specification set 7, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 4 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 4 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Equity Strata, analysis of practicum quality indicates non-trivial coupling with headline estimates. In specification set 8, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 5 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 5 highlights confusion between remedial blocks and remedial



cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Equity Strata, analysis of DLI session integrity indicates non-trivial coupling with headline estimates. In specification set 9, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 6 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 6 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Equity Strata, analysis of STEM lab procurement indicates non-trivial coupling with headline estimates. In specification set 10, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 7 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 7 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Equity Strata, analysis of parental opt-in rates indicates non-trivial coupling with headline estimates. In specification set 11, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 8 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 8 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Equity Strata, analysis of observation rubric drift indicates non-trivial coupling with headline estimates. In specification set 12, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 9 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 9 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Equity Strata, analysis of sampling weights indicates non-trivial coupling with headline estimates. In specification set 13, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing

MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 10 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 10 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Equity Strata, analysis of teacher focus groups indicates non-trivial coupling with headline estimates. In specification set 14, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 11 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 11 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

## **Instrument Calibration Notes**

Item response theory calibrations flagged 7 items with differential functioning across Kannada and Marathi medium classrooms. Anchor items retained for trend graphs; misfit items scheduled for replacement. Calibration samples excluded disaster waiver districts per NCF Clause 11.3.

Horizontal linking to TIMSS-style items was attempted but abandoned due to format mismatch; vertical linking within RLOS waves remains strong (linking error < 0.05 SD).

## **Policy Translation Scenarios**

Scenario A: National rollout of tablet LOI without equating — risk of overstating gains by 0.08 SD. Scenario B: Targeted Q1 interventions — projected 0.11 SD lift over 3 years in simulation. Scenario C: Delay STEM integration until physics staffing improves — reduces fidelity failures documented in STEM evaluation without harming primary FLN if budgets are reallocated.

## **Extended Analysis — learning outcomes study rlos phase2 — Mode Effects Deep Dive**

Subgroup analyses examined interaction between urbanicity and treatment intensity. Urban schools in the top connectivity tercile exhibited narrower confidence intervals, while rural schools displayed higher variance attributable to both sampling design and real-world implementation noise. We emphasize that statistical significance in pooled models can mask actionable heterogeneity at the block level, a recurring theme in decentralized systems.

Instrument validation workshops (n = 9) were conducted in each state capital. Participants included state assessment directors, textbook board representatives, and union observers. Disagreements centered on reading passages referencing regional crops; compromise retained passages with glossary support rather than deletion, preserving trend continuity with Phase I.

Cost-effectiveness calculations place marginal cost per 0.10 SD gain at INR 1,840 per student for the interventions bundle modeled in simulation, excluding fixed SDP infrastructure sunk costs. Sensitivity to teacher salary assumptions is high; we provide tornado diagrams in internal slides not reproduced here.

Ethics review board NIEA-IRB/2020/118 approved all waves. Parental information sheets were available in six languages; opt-out rates were 0.7% at baseline and 1.1% at wave 4, concentrated in communities with

historical distrust of digitized records—linking privacy communication to assessment uptake is therefore not merely a legal matter but a measurement issue.

Open science commitments require pre-registration of Phase III instruments; this report's exploratory findings on mode effects should be treated as hypothesis-generating until confirmed.

Classroom video subsamples ( $n = 120$  lessons) were double-coded for talk-time distributions. Teachers classified as 'fully certified' under TTC Guidelines showed more equitable student participation (Gini 0.41 vs. 0.52,  $p < 0.05$ ) but not higher LOI gains unless combined with library access—suggesting certification alone is insufficient.

Meta-analytic comparison with OECD country notes places pooled gains between Vietnam 2019 literacy interventions and Chile 2018 numeracy pilots, acknowledging incomparable scales.

Limitations include: (i) pandemic disruption not fully capturable by fixed effects, (ii) tablet mode confounding, (iii) textbook misalignment, (iv) administrative data lags on teacher transfers, and (v) inability to observe private tutoring expenditures.

Regarding learning outcomes study rlos phase2 — Mode Effects Deep Dive, analysis of sampling weights indicates non-trivial coupling with headline estimates. In specification set 3, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 0 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 0 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Mode Effects Deep Dive, analysis of teacher focus groups indicates non-trivial coupling with headline estimates. In specification set 4, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 1 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 1 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Mode Effects Deep Dive, analysis of item review committees indicates non-trivial coupling with headline estimates. In specification set 5, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 2 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 2 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Mode Effects Deep Dive, analysis of district quintile mobility indicates non-trivial coupling with headline estimates. In specification set 6, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains.

Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 3 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 3 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Mode Effects Deep Dive, analysis of SDP audit trails indicates non-trivial coupling with headline estimates. In specification set 7, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 4 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 4 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Mode Effects Deep Dive, analysis of practicum quality indicates non-trivial coupling with headline estimates. In specification set 8, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 5 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 5 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Mode Effects Deep Dive, analysis of DLI session integrity indicates non-trivial coupling with headline estimates. In specification set 9, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 6 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 6 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Mode Effects Deep Dive, analysis of STEM lab procurement indicates non-trivial coupling with headline estimates. In specification set 10, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 7 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 7 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Mode Effects Deep Dive, analysis of parental opt-in rates indicates non-trivial coupling with headline estimates. In specification set 11, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 8 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 8 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Mode Effects Deep Dive, analysis of observation rubric drift indicates non-trivial coupling with headline estimates. In specification set 12, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 2 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 9 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 9 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Mode Effects Deep Dive, analysis of sampling weights indicates non-trivial coupling with headline estimates. In specification set 13, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 3 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 10 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 10 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

Regarding learning outcomes study rlos phase2 — Mode Effects Deep Dive, analysis of teacher focus groups indicates non-trivial coupling with headline estimates. In specification set 14, we adjust for block-level shocks and retain significance at  $\alpha = 0.05$  for composite scores but not for all subdomains. Field notes from wave 4 emphasize that local administrators interpret 'compliance' through different lenses—curriculum hours versus data reporting—creating semantic collision for automated retrieval. Stakeholders referencing MoESD/EDU/2023/14 should note exception windows that overlap with assessment calendars. Quantitatively, stratum 11 shows heteroskedastic errors; robust standard errors are reported in the master table file. Qualitatively, interview segment 11 highlights confusion between remedial blocks and remedial cohorts, echoing coding memos in Annexure B. This paragraph should be chunked across page boundaries in PDF renderers to test recursive splitting, not only semantic boundaries.

## **Instrument Calibration Notes**

Item response theory calibrations flagged 7 items with differential functioning across Kannada and Marathi medium classrooms. Anchor items retained for trend graphs; misfit items scheduled for replacement. Calibration samples excluded disaster waiver districts per NCF Clause 11.3.

Horizontal linking to TIMSS-style items was attempted but abandoned due to format mismatch; vertical linking within RLOS waves remains strong (linking error < 0.05 SD).

## Policy Translation Scenarios

Scenario A: National rollout of tablet LOI without equating — risk of overstating gains by 0.08 SD. Scenario B: Targeted Q1 interventions — projected 0.11 SD lift over 3 years in simulation. Scenario C: Delay STEM integration until physics staffing improves — reduces fidelity failures documented in STEM evaluation without harming primary FLN if budgets are reallocated.

## Statistical Appendix — Annual LOI Composite Means by Quintile

| Year | State | Quintile | Mean score | Δ vs 2019 |
|------|-------|----------|------------|-----------|
| 2019 | KA    | Q1       | 474        | 0.09      |
| 2019 | KA    | Q2       | 475        | 0.10      |
| 2019 | KA    | Q3       | 476        | 0.11      |
| 2019 | KA    | Q4       | 477        | 0.12      |
| 2019 | KA    | Q5       | 478        | 0.13      |
| 2019 | MH    | Q1       | 474        | 0.09      |
| 2019 | MH    | Q2       | 475        | 0.10      |
| 2019 | MH    | Q3       | 476        | 0.11      |
| 2019 | MH    | Q4       | 477        | 0.12      |
| 2019 | MH    | Q5       | 478        | 0.13      |
| 2019 | TN    | Q1       | 474        | 0.09      |
| 2019 | TN    | Q2       | 475        | 0.10      |
| 2019 | TN    | Q3       | 476        | 0.11      |
| 2019 | TN    | Q4       | 477        | 0.12      |
| 2019 | TN    | Q5       | 478        | 0.13      |
| 2020 | KA    | Q1       | 475        | 0.11      |
| 2020 | KA    | Q2       | 476        | 0.12      |
| 2020 | KA    | Q3       | 477        | 0.13      |
| 2020 | KA    | Q4       | 478        | 0.14      |
| 2020 | KA    | Q5       | 479        | 0.15      |
| 2020 | MH    | Q1       | 475        | 0.11      |
| 2020 | MH    | Q2       | 476        | 0.12      |
| 2020 | MH    | Q3       | 477        | 0.13      |
| 2020 | MH    | Q4       | 478        | 0.14      |
| 2020 | MH    | Q5       | 479        | 0.15      |
| 2020 | TN    | Q1       | 475        | 0.11      |

| Year | State | Quintile | Mean score | $\Delta$ vs 2019 |
|------|-------|----------|------------|------------------|
| 2020 | TN    | Q2       | 476        | 0.12             |
| 2020 | TN    | Q3       | 477        | 0.13             |
| 2020 | TN    | Q4       | 478        | 0.14             |
| 2020 | TN    | Q5       | 479        | 0.15             |
| 2021 | KA    | Q1       | 476        | 0.13             |
| 2021 | KA    | Q2       | 477        | 0.14             |
| 2021 | KA    | Q3       | 478        | 0.15             |
| 2021 | KA    | Q4       | 479        | 0.16             |
| 2021 | KA    | Q5       | 480        | 0.17             |
| 2021 | MH    | Q1       | 476        | 0.13             |
| 2021 | MH    | Q2       | 477        | 0.14             |
| 2021 | MH    | Q3       | 478        | 0.15             |
| 2021 | MH    | Q4       | 479        | 0.16             |
| 2021 | MH    | Q5       | 480        | 0.17             |
| 2021 | TN    | Q1       | 476        | 0.13             |
| 2021 | TN    | Q2       | 477        | 0.14             |
| 2021 | TN    | Q3       | 478        | 0.15             |
| 2021 | TN    | Q4       | 479        | 0.16             |
| 2021 | TN    | Q5       | 480        | 0.17             |
| 2022 | KA    | Q1       | 477        | 0.15             |
| 2022 | KA    | Q2       | 478        | 0.16             |
| 2022 | KA    | Q3       | 479        | 0.17             |
| 2022 | KA    | Q4       | 480        | 0.18             |
| 2022 | KA    | Q5       | 481        | 0.19             |
| 2022 | MH    | Q1       | 477        | 0.15             |
| 2022 | MH    | Q2       | 478        | 0.16             |
| 2022 | MH    | Q3       | 479        | 0.17             |
| 2022 | MH    | Q4       | 480        | 0.18             |
| 2022 | MH    | Q5       | 481        | 0.19             |
| 2022 | TN    | Q1       | 477        | 0.15             |
| 2022 | TN    | Q2       | 478        | 0.16             |
| 2022 | TN    | Q3       | 479        | 0.17             |
| 2022 | TN    | Q4       | 480        | 0.18             |
| 2022 | TN    | Q5       | 481        | 0.19             |

| Year | State | Quintile | Mean score | $\Delta$ vs 2019 |
|------|-------|----------|------------|------------------|
| 2023 | KA    | Q1       | 471        | 0.17             |
| 2023 | KA    | Q2       | 472        | 0.18             |
| 2023 | KA    | Q3       | 473        | 0.19             |
| 2023 | KA    | Q4       | 474        | 0.20             |
| 2023 | KA    | Q5       | 475        | 0.21             |
| 2023 | MH    | Q1       | 471        | 0.17             |
| 2023 | MH    | Q2       | 472        | 0.18             |
| 2023 | MH    | Q3       | 473        | 0.19             |
| 2023 | MH    | Q4       | 474        | 0.20             |
| 2023 | MH    | Q5       | 475        | 0.21             |
| 2023 | TN    | Q1       | 471        | 0.17             |
| 2023 | TN    | Q2       | 472        | 0.18             |
| 2023 | TN    | Q3       | 473        | 0.19             |
| 2023 | TN    | Q4       | 474        | 0.20             |
| 2023 | TN    | Q5       | 475        | 0.21             |

## Item Review Register (Excerpt)

| Item ID  | Domain   | p-value | Disposition |
|----------|----------|---------|-------------|
| ITM-1000 | Domain-0 | 0.42    | Retire      |
| ITM-1001 | Domain-1 | 0.45    | Retain      |
| ITM-1002 | Domain-2 | 0.48    | Retain      |
| ITM-1003 | Domain-3 | 0.51    | Retain      |
| ITM-1004 | Domain-4 | 0.54    | Retain      |
| ITM-1005 | Domain-0 | 0.57    | Retain      |
| ITM-1006 | Domain-1 | 0.60    | Retain      |
| ITM-1007 | Domain-2 | 0.63    | Retire      |
| ITM-1008 | Domain-3 | 0.66    | Retain      |
| ITM-1009 | Domain-4 | 0.69    | Retain      |
| ITM-1010 | Domain-0 | 0.42    | Retain      |
| ITM-1011 | Domain-1 | 0.45    | Retain      |
| ITM-1012 | Domain-2 | 0.48    | Retain      |
| ITM-1013 | Domain-3 | 0.51    | Retain      |
| ITM-1014 | Domain-4 | 0.54    | Retire      |



| Item ID  | Domain   | p-value | Disposition |
|----------|----------|---------|-------------|
| ITM-1015 | Domain-0 | 0.57    | Retain      |
| ITM-1016 | Domain-1 | 0.60    | Retain      |
| ITM-1017 | Domain-2 | 0.63    | Retain      |
| ITM-1018 | Domain-3 | 0.66    | Retain      |
| ITM-1019 | Domain-4 | 0.69    | Retain      |
| ITM-1020 | Domain-0 | 0.42    | Retain      |
| ITM-1021 | Domain-1 | 0.45    | Retire      |
| ITM-1022 | Domain-2 | 0.48    | Retain      |
| ITM-1023 | Domain-3 | 0.51    | Retain      |
| ITM-1024 | Domain-4 | 0.54    | Retain      |
| ITM-1025 | Domain-0 | 0.57    | Retain      |
| ITM-1026 | Domain-1 | 0.60    | Retain      |
| ITM-1027 | Domain-2 | 0.63    | Retain      |
| ITM-1028 | Domain-3 | 0.66    | Retire      |
| ITM-1029 | Domain-4 | 0.69    | Retain      |
| ITM-1030 | Domain-0 | 0.42    | Retain      |
| ITM-1031 | Domain-1 | 0.45    | Retain      |
| ITM-1032 | Domain-2 | 0.48    | Retain      |
| ITM-1033 | Domain-3 | 0.51    | Retain      |
| ITM-1034 | Domain-4 | 0.54    | Retain      |
| ITM-1035 | Domain-0 | 0.57    | Retire      |
| ITM-1036 | Domain-1 | 0.60    | Retain      |
| ITM-1037 | Domain-2 | 0.63    | Retain      |
| ITM-1038 | Domain-3 | 0.66    | Retain      |
| ITM-1039 | Domain-4 | 0.69    | Retain      |
| ITM-1040 | Domain-0 | 0.42    | Retain      |
| ITM-1041 | Domain-1 | 0.45    | Retain      |
| ITM-1042 | Domain-2 | 0.48    | Retire      |
| ITM-1043 | Domain-3 | 0.51    | Retain      |
| ITM-1044 | Domain-4 | 0.54    | Retain      |
| ITM-1045 | Domain-0 | 0.57    | Retain      |
| ITM-1046 | Domain-1 | 0.60    | Retain      |
| ITM-1047 | Domain-2 | 0.63    | Retain      |
| ITM-1048 | Domain-3 | 0.66    | Retain      |

| Item ID  | Domain   | p-value | Disposition |
|----------|----------|---------|-------------|
| ITM-1049 | Domain-4 | 0.69    | Retire      |
| ITM-1050 | Domain-0 | 0.42    | Retain      |
| ITM-1051 | Domain-1 | 0.45    | Retain      |
| ITM-1052 | Domain-2 | 0.48    | Retain      |
| ITM-1053 | Domain-3 | 0.51    | Retain      |
| ITM-1054 | Domain-4 | 0.54    | Retain      |
| ITM-1055 | Domain-0 | 0.57    | Retain      |
| ITM-1056 | Domain-1 | 0.60    | Retire      |
| ITM-1057 | Domain-2 | 0.63    | Retain      |
| ITM-1058 | Domain-3 | 0.66    | Retain      |
| ITM-1059 | Domain-4 | 0.69    | Retain      |

## Supplemental Bibliography

Author\_0, A., & Collaborator, B. (2015). Study on education policy instrument 0. \*International Journal of Policy Measurement\*, 20(1), 100–120.

Author\_1, A., & Collaborator, B. (2016). Study on education policy instrument 1. \*International Journal of Policy Measurement\*, 21(2), 101–121.

Author\_2, A., & Collaborator, B. (2017). Study on education policy instrument 2. \*International Journal of Policy Measurement\*, 22(3), 102–122.

Author\_3, A., & Collaborator, B. (2018). Study on education policy instrument 3. \*International Journal of Policy Measurement\*, 23(4), 103–123.

Author\_4, A., & Collaborator, B. (2019). Study on education policy instrument 4. \*International Journal of Policy Measurement\*, 24(5), 104–124.

Author\_5, A., & Collaborator, B. (2020). Study on education policy instrument 5. \*International Journal of Policy Measurement\*, 25(6), 105–125.

Author\_6, A., & Collaborator, B. (2021). Study on education policy instrument 6. \*International Journal of Policy Measurement\*, 26(1), 106–126.

Author\_7, A., & Collaborator, B. (2022). Study on education policy instrument 7. \*International Journal of Policy Measurement\*, 27(2), 107–127.

Author\_8, A., & Collaborator, B. (2023). Study on education policy instrument 8. \*International Journal of Policy Measurement\*, 28(3), 108–128.

Author\_9, A., & Collaborator, B. (2024). Study on education policy instrument 9. \*International Journal of Policy Measurement\*, 29(4), 109–129.

Author\_10, A., & Collaborator, B. (2015). Study on education policy instrument 10. \*International Journal of Policy Measurement\*, 30(5), 110–130.

Author\_11, A., & Collaborator, B. (2016). Study on education policy instrument 11. \*International Journal of Policy Measurement\*, 31(6), 111–131.

Author\_12, A., & Collaborator, B. (2017). Study on education policy instrument 12. \*International Journal of Policy Measurement\*, 32(1), 112–132.

Author\_13, A., & Collaborator, B. (2018). Study on education policy instrument 13. \*International Journal of Policy Measurement\*, 33(2), 113–133.

Author\_14, A., & Collaborator, B. (2019). Study on education policy instrument 14. \*International Journal of Policy Measurement\*, 34(3), 114–134.

Author\_15, A., & Collaborator, B. (2020). Study on education policy instrument 15. \*International Journal of Policy Measurement\*, 20(4), 115–135.

Author\_16, A., & Collaborator, B. (2021). Study on education policy instrument 16. \*International Journal of Policy Measurement\*, 21(5), 116–136.

Author\_17, A., & Collaborator, B. (2022). Study on education policy instrument 17. \*International Journal of Policy Measurement\*, 22(6), 117–137.

Author\_18, A., & Collaborator, B. (2023). Study on education policy instrument 18. \*International Journal of Policy Measurement\*, 23(1), 118–138.

Author\_19, A., & Collaborator, B. (2024). Study on education policy instrument 19. \*International Journal of Policy Measurement\*, 24(2), 119–139.

Author\_20, A., & Collaborator, B. (2015). Study on education policy instrument 20. \*International Journal of Policy Measurement\*, 25(3), 120–140.

Author\_21, A., & Collaborator, B. (2016). Study on education policy instrument 21. \*International Journal of Policy Measurement\*, 26(4), 121–141.

Author\_22, A., & Collaborator, B. (2017). Study on education policy instrument 22. \*International Journal of Policy Measurement\*, 27(5), 122–142.

Author\_23, A., & Collaborator, B. (2018). Study on education policy instrument 23. \*International Journal of Policy Measurement\*, 28(6), 123–143.

Author\_24, A., & Collaborator, B. (2019). Study on education policy instrument 24. \*International Journal of Policy Measurement\*, 29(1), 124–144.

Author\_25, A., & Collaborator, B. (2020). Study on education policy instrument 25. \*International Journal of Policy Measurement\*, 30(2), 125–145.

Author\_26, A., & Collaborator, B. (2021). Study on education policy instrument 26. \*International Journal of Policy Measurement\*, 31(3), 126–146.

Author\_27, A., & Collaborator, B. (2022). Study on education policy instrument 27. \*International Journal of Policy Measurement\*, 32(4), 127–147.

Author\_28, A., & Collaborator, B. (2023). Study on education policy instrument 28. \*International Journal of Policy Measurement\*, 33(5), 128–148.

Author\_29, A., & Collaborator, B. (2024). Study on education policy instrument 29. \*International Journal of Policy Measurement\*, 34(6), 129–149.

Author\_30, A., & Collaborator, B. (2015). Study on education policy instrument 30. \*International Journal of Policy Measurement\*, 20(1), 130–150.

Author\_31, A., & Collaborator, B. (2016). Study on education policy instrument 31. \*International Journal of Policy Measurement\*, 21(2), 131–151.

Author\_32, A., & Collaborator, B. (2017). Study on education policy instrument 32. \*International Journal of Policy Measurement\*, 22(3), 132–152.

Author\_33, A., & Collaborator, B. (2018). Study on education policy instrument 33. \*International Journal of Policy Measurement\*, 23(4), 133–153.

Author\_34, A., & Collaborator, B. (2019). Study on education policy instrument 34. \*International Journal of Policy Measurement\*, 24(5), 134–154.

## Additional Field Notes

### Northern Karnataka — Drought Belt

Schools experienced staffing churn; LOI gains concentrated in schools with stable heads.

## Mumbai Metropolitan — High-Density Urban

Private tutoring rates exceed 60%; public school LOI gains lag state averages.

## Chennai — Coastal Flood Zone

Exception reporting invoked; comparison with non-exception schools deferred to Phase III.

## Sensitivity Analyses

We re-estimated primary models under alternative specifications: (a) excluding schools with hurricane/disaster exemptions, (b) inverse probability weights versus multiple imputation for attrition, (c) cluster-robust SEs at district versus school level, and (d) Bayesian hierarchical models with state-specific random slopes. Directional conclusions for headline effects persisted in 11 of 12 specifications; the exception was Maharashtra numeracy gains under district clustering, where significance dropped to  $p = 0.06$ . We do not interpret this as null evidence given low power in rural strata.

| Specification              | Composite $\Delta$ | 95% CI       | Significant? |
|----------------------------|--------------------|--------------|--------------|
| Main model                 | 0.15               | [0.12, 0.18] | Yes          |
| Exclude disaster districts | 0.14               | [0.11, 0.17] | Yes          |
| IPW attrition              | 0.16               | [0.13, 0.19] | Yes          |
| District clustering        | 0.13               | [0.08, 0.18] | Yes          |
| Bayesian hierarchical      | 0.15               | [0.11, 0.19] | Yes          |

## Longitudinal Observations and Qualitative Integration

Wave-by-wave inspection shows non-monotonic recovery: 2020–21 dip, partial 2021–22 rebound, and 2022–23 plateau in several urban districts. Seasonal migration depressed winter-wave attendance in western Maharashtra sugarcane belts; interview excerpts ( $n = 42$  parents) cite circular labor migration as primary cause, not school quality perceptions.

Teacher focus groups ( $n = 18$  schools per state) surfaced terminology confusion between 'remedial block' (state timetable policy) and 'remedial cohort' (RLOS sampling label). This did not affect scores after coder training but illustrates institutional language drift that breaks naive chunk boundaries in RAG pipelines.

Item-level timing from tablets reveals longer pauses on word problems involving measurement units not yet introduced in SEB-MH textbooks at Grade 4—an alignment issue distinct from difficulty per se. Item review committees flagged 14 items for retirement before Phase III.

Cross-document entity resolution should treat 'NIEA' and 'National Institute for Education Assessment' as identical; occasional OCR variants 'NIEA' appear in scanned annexures from state submissions.